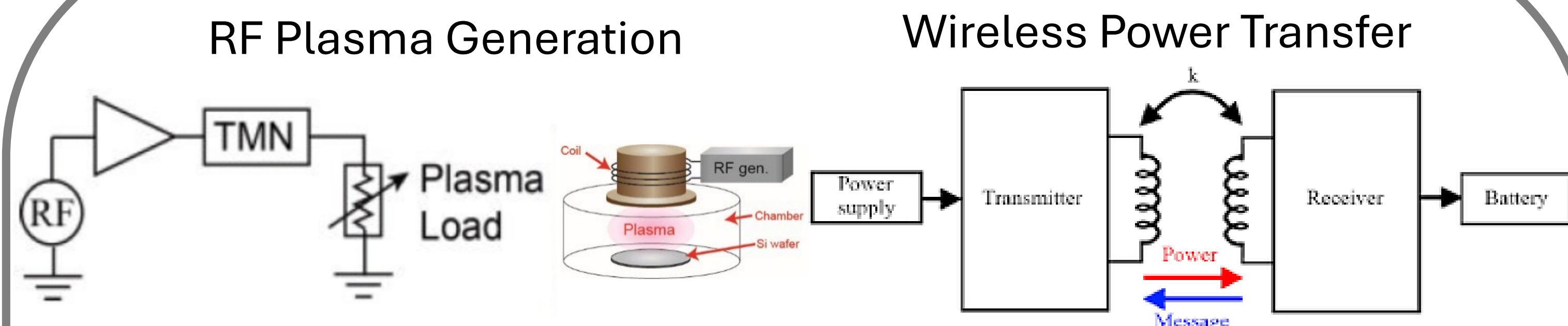


High-Frequency Applications



State-of-the-art Switched-Mode Power Amplifiers [1-3]

- High efficiency ✓
- Wide load & power range ✓
- HF (3-30 MHz) & VHF (30-300 MHz) ✗ Limited to 13.56 MHz

Many SMPA architectures use high-side / floating switches (e.g., Class-D)

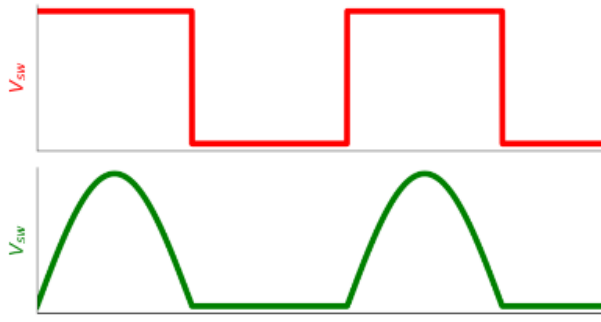
✗ Increased gate-drive complexity and frequency limits from level shifters [4]

✓ Ground referenced switch topologies (e.g., Class-E)

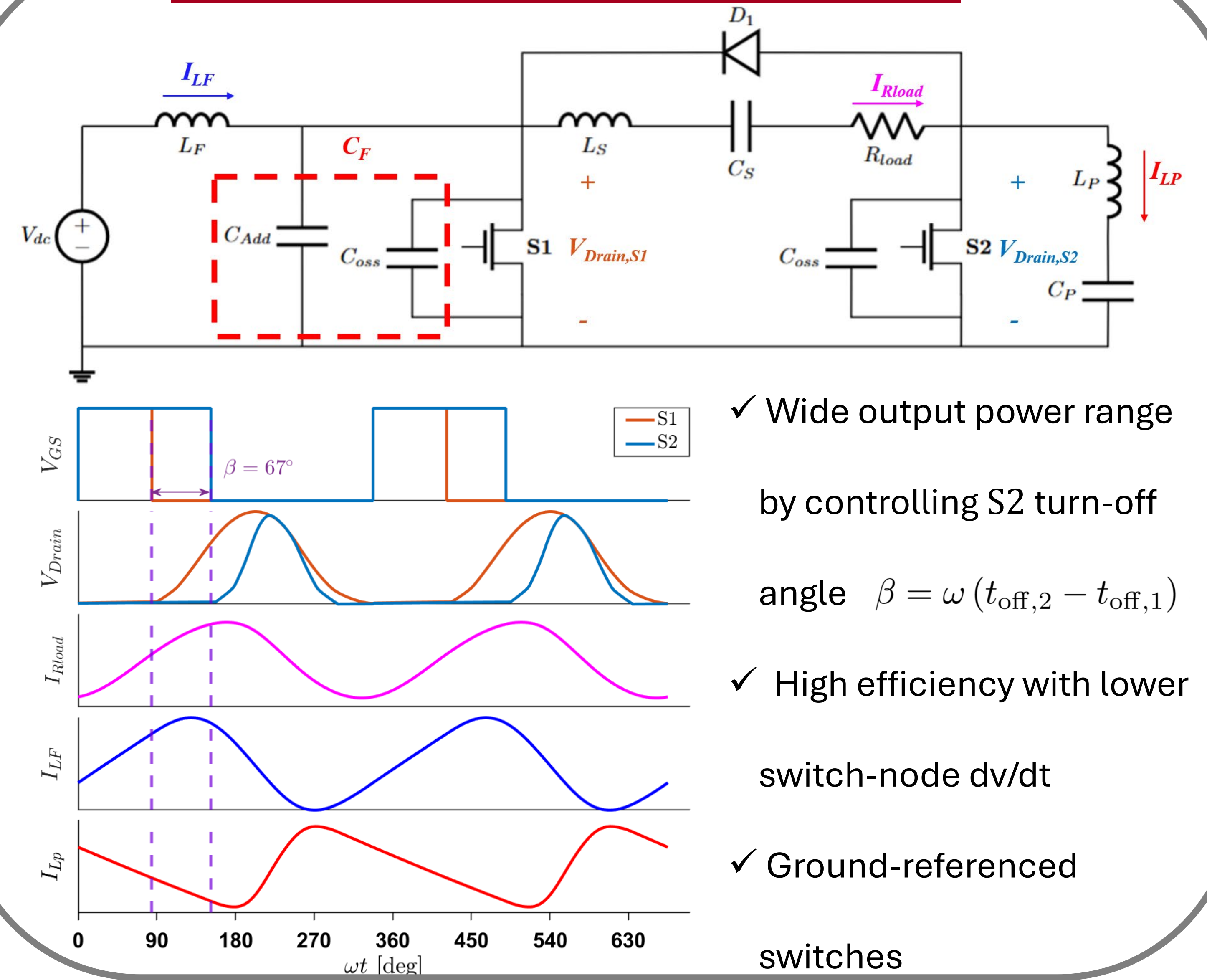
Frequency-dependent device losses become dominant and limit performance

✗ Device losses increase with switch-node dv/dt [5,6]

✓ Lower switch-node dv/dt in Class-E vs Class-D [7]



Class-E PA with β Modulation

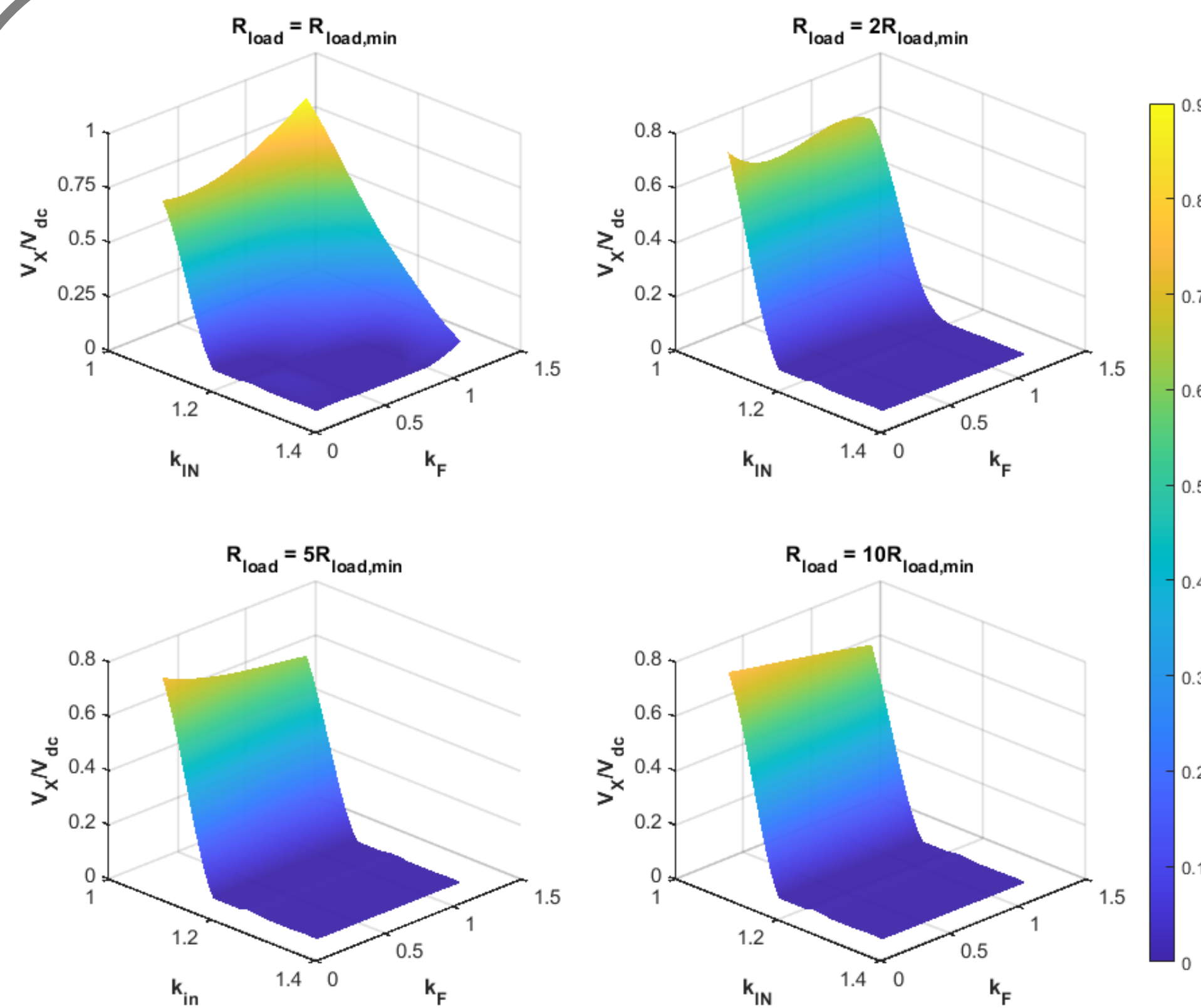


✓ Wide output power range
by controlling S2 turn-off
angle $\beta = \omega(t_{\text{off},2} - t_{\text{off},1})$

✓ High efficiency with lower
switch-node dv/dt

✓ Ground-referenced
switches

Design Methodology



Selected: $k_{IN} = 1.15$, $k_F = 0.8$

Input-network design

$$k_{IN} = \frac{1}{2\pi f_{sw} \sqrt{L_F C_F}}$$

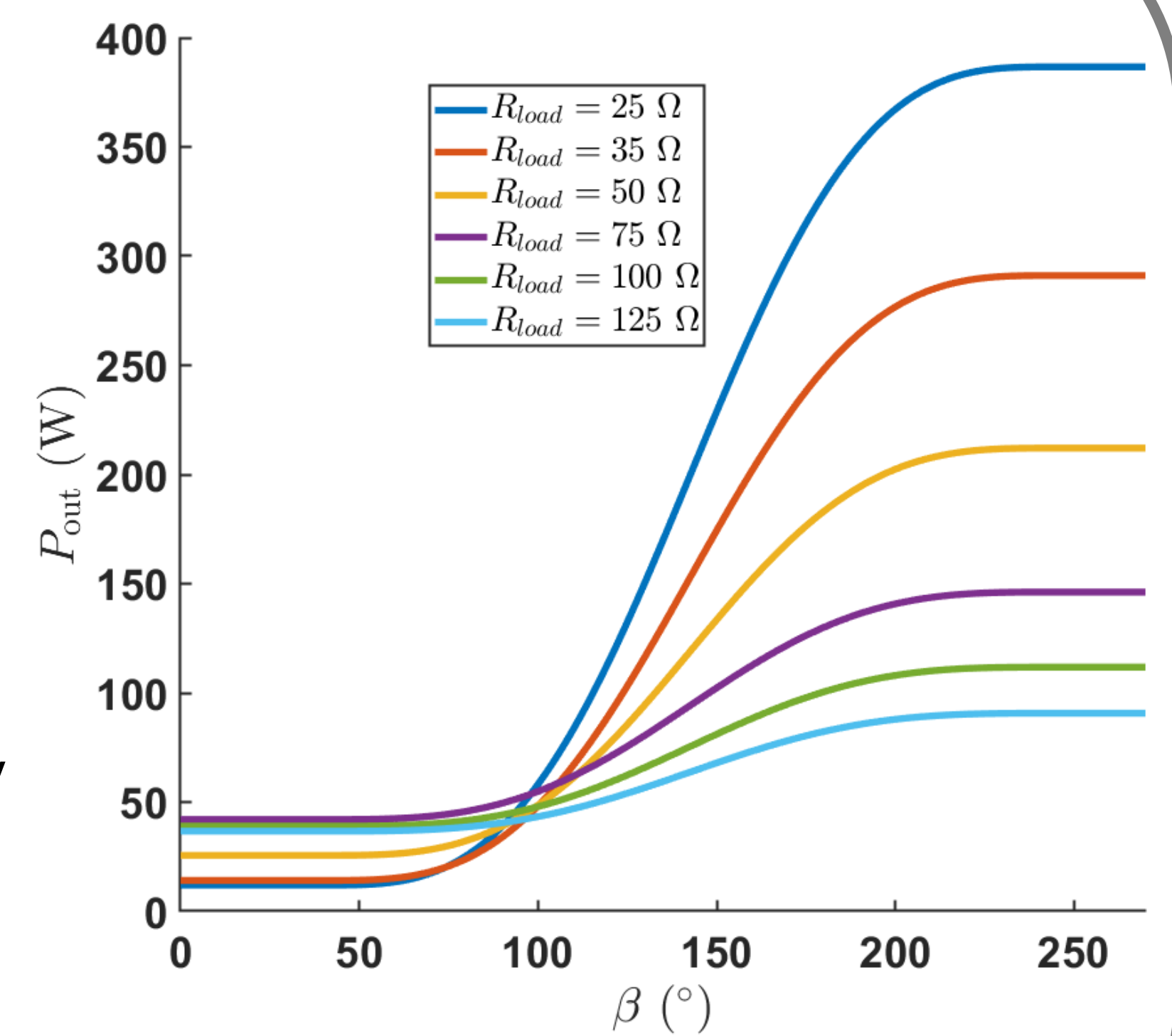
$$k_F = \frac{\sqrt{L_F / C_F}}{R_{load,min}}$$

- Numerical waveform analysis maps k_{IN} - k_F region that maintains soft switching across wide load range [8]

Loss optimization

$$P_{loss} = P_{cond,S1} + P_{C_{oss},S1} + P_{cond,S2} + P_{C_{oss},S2} + P_{L_f} + P_{L_s} + P_{L_p}$$

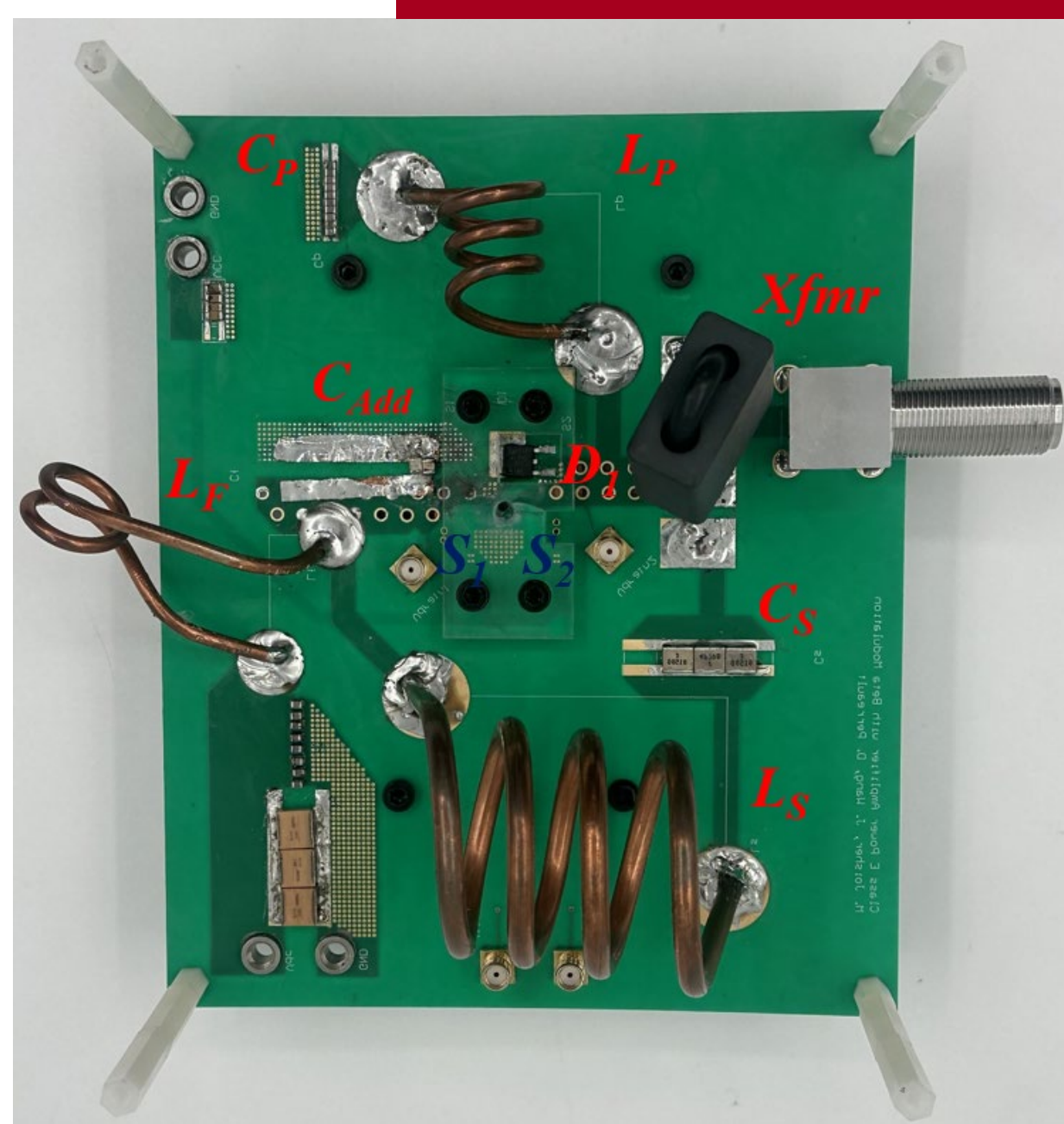
- Device and component values optimized using experimentally measured loss data [7]
- 25% on-duty for S1 to minimize C_{oss} loss



Simulation results :

Wide load and power range

Hardware Prototype



Parameter	Value
f_{sw}	30.9 MHz
V_{dc}	160 V (maximum β) 130 V (minimum β)
L_F	85 nH
C_{Add}	198 pF
L_S	668 nH
C_S	51.7 pF
L_P	118 nH
C_P	150 nF
S_1, S_2	GS66506T
D_1	STPSC406B
D_{S1}	25%
D_{S2}	25-100%
R_{load}	25 - 50 Ω
Transformer winding	1:1 RG-58 coax
Transformer core	Fair-Rite 67

β Modulation Experimental Results

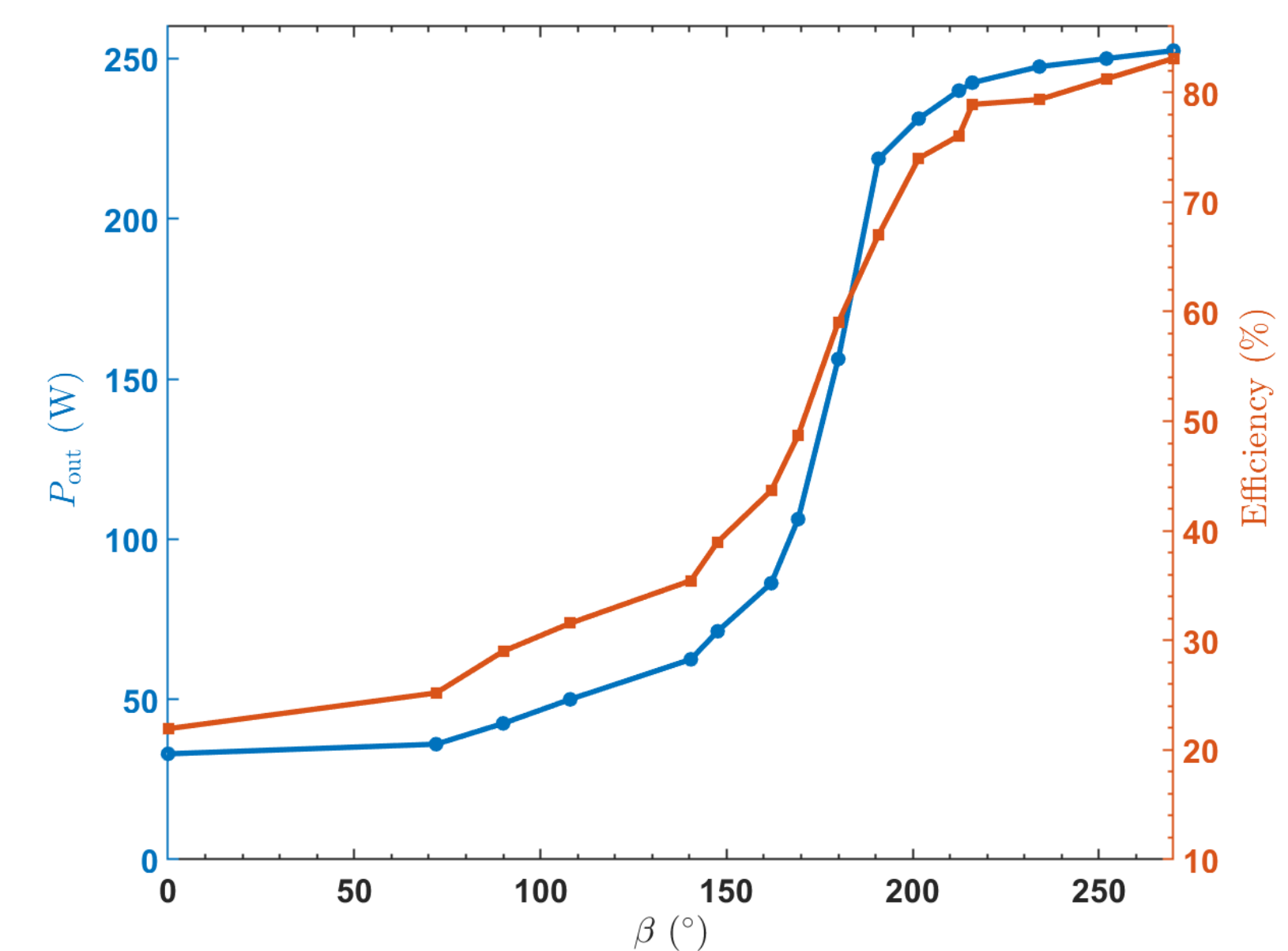
$V_{dc} = 130$ V, $R_{load} = 25$ Ω ,

$\beta = 0^\circ - 270^\circ$

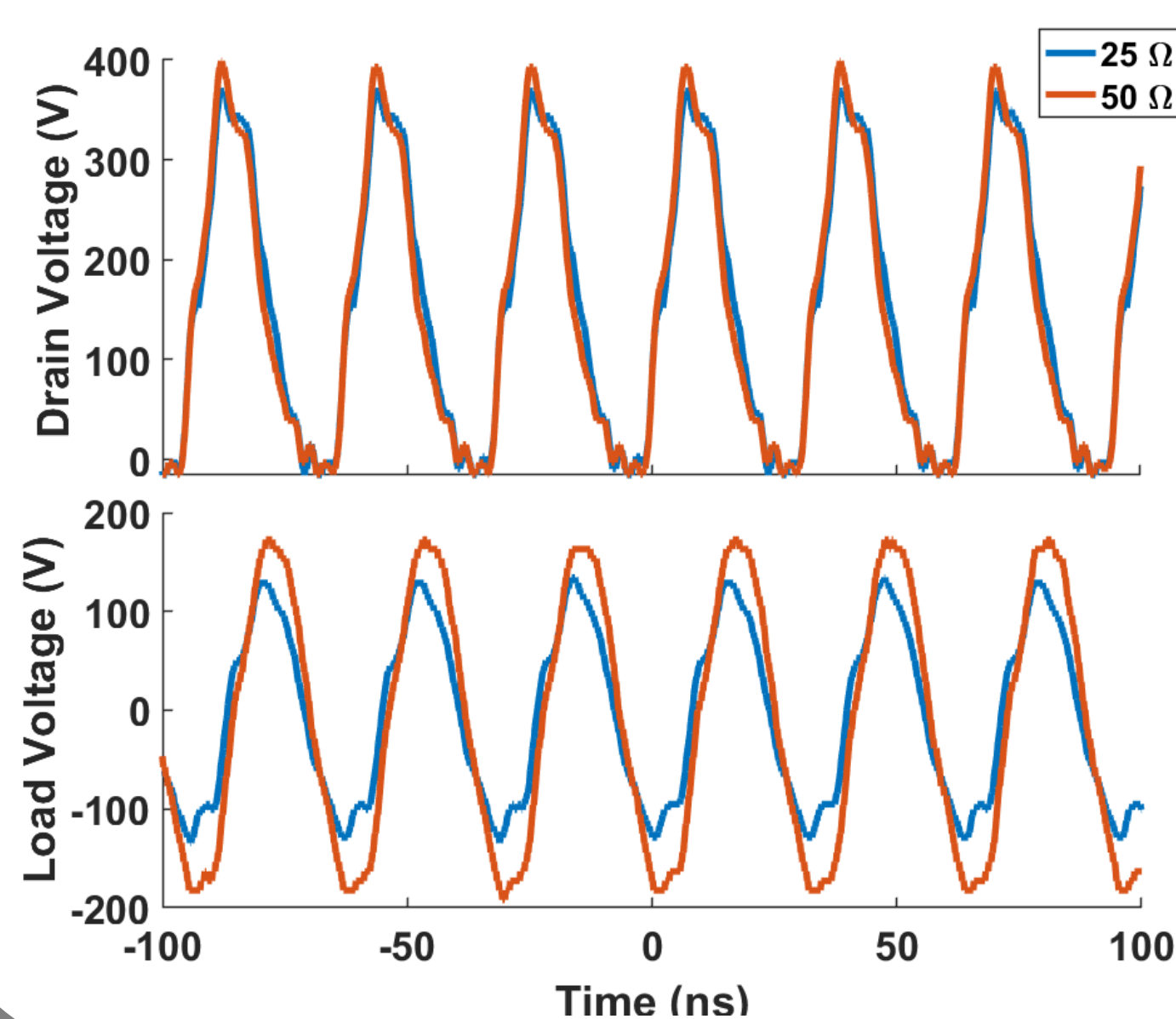
Maximum β : **250 W at 83%**

Minimum β : **33 W at 22%**

Wide output power range: **8.3×**



Maximum β Experimental Results



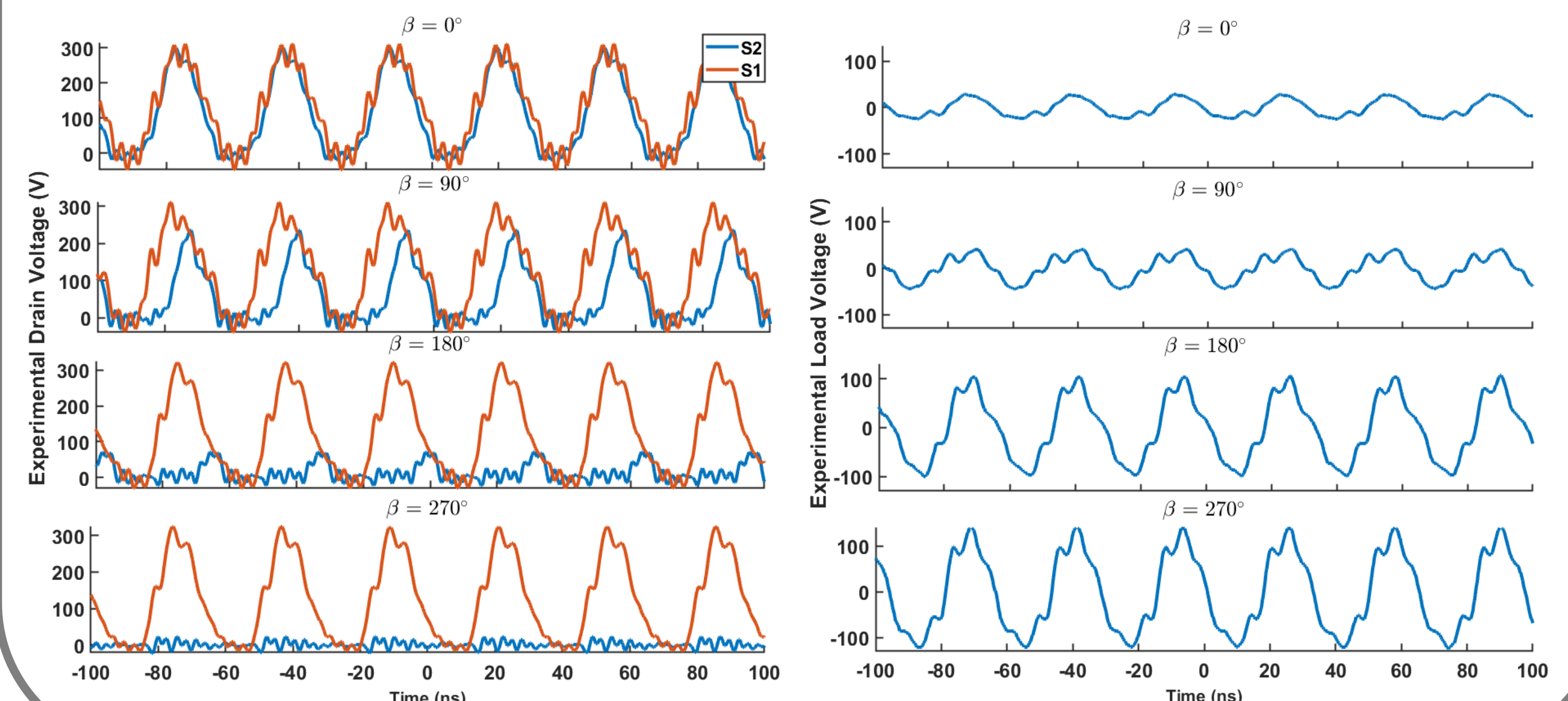
$V_{dc} = 160$ V,

$\beta = 270^\circ$ (S2 always ON),

$R_{load} = 25$ Ω : **350 W at 84.3%**

$R_{load} = 50$ Ω : **228 W at 81%**

Soft-switching maintained for S1



[1] H. Zhang et al., "High-Efficiency Wide-Range RF Power Generation Systems With Discrete Power Back-Off From Multiple Inverters," IEEE TPEL 2025

[2] X. Zan et al., "Wide-Range Switched-Mode Power Amplifier Architecture," IEEE COMPEL 2023

[3] T. Oladugba et al., "Wide-Range Current-Mode Class-D Power Amplifier," IEEE COMPEL 2025

[4] C. Zhang, S. Zhao, Z. Sun, and R. S. Y. Hui, "A Self-Triggering ZVS Gate Drive for Multi-MHz Class-D Inverters," IEEE APEC 2026

[5] G. Zulauf et al., "COSS Losses in 600 V GaN Power Semiconductors in Soft-Switched, High- and very-High-Frequency Power Converters," IEEE TPEL 2018

[6] A. Garg et al., "Characterizing High-Frequency GaN HEMT Coss Losses with Class-D Waveforms," IEEE COMPEL 2026

[7] M. N. V. Joisher et al., "High-Frequency Characterization of Coss Loss and Dynamic RdsON in GaN," IEEE APEC 2026

[8] L. Roslaniec et al., "Design of Single-Switch Inverters for Variable Resistance/Load Modulation Operation," IEEE TPEL 2015